

SMART INDIA HACKATHON 2026



- **Problem Statement ID – 26047**
- **Problem Statement Title- Patient Case-Taking Software**
- **Theme- Medtech/Biotech**
- **PS Category- Software**
- **Team Name: The Black Hats**

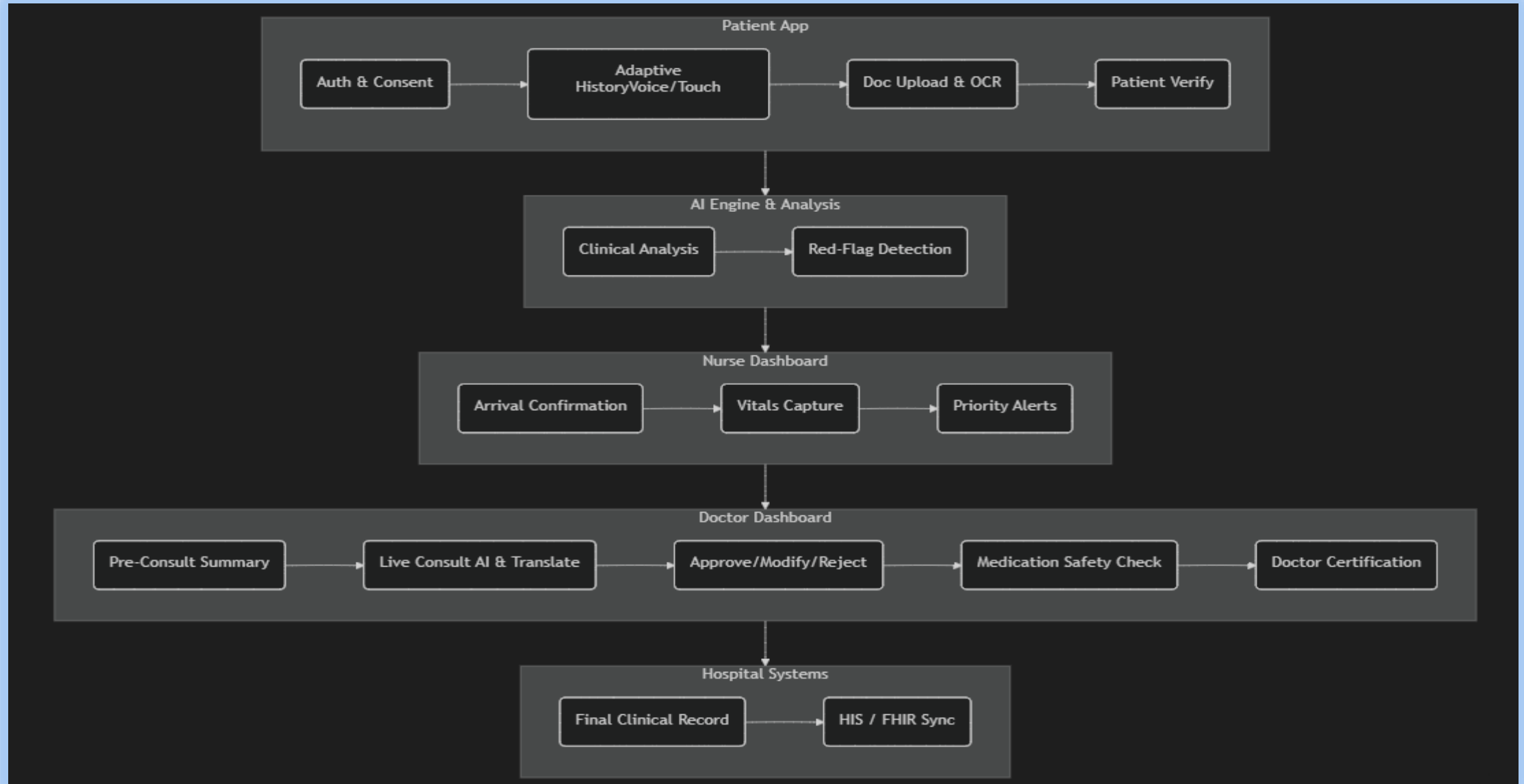


The Problem

- Doctors spend valuable consultation time collecting basic history.
- Patients may struggle with language, recall and repeated questioning.
- Prescriptions, reports and previous records are scattered across documents and systems.
- Important clinical information can be missed or remain unstructured.

Our Solution

- **AI History Taking:** Multilingual voice/touch interaction with adaptive, complaint-specific questions.
- **Medical Record Digitization:** Scans prescriptions and reports, including handwritten documents, and extracts useful clinical information.
- **Clinical History Generation:** Combines current responses with past records to create a structured, chronological, doctor-ready history.
- **Clinical Support:** Highlights red flags, important abnormalities and potential interactions for **doctor review**, not autonomous diagnosis.



Multi-model AI Orchestration:

- Employs dynamic switching, fallback consensus, and an **Evidence Engine** to ensure AI outputs are linked to trusted sources.

ABDM / ABHA / FHIR Integration:

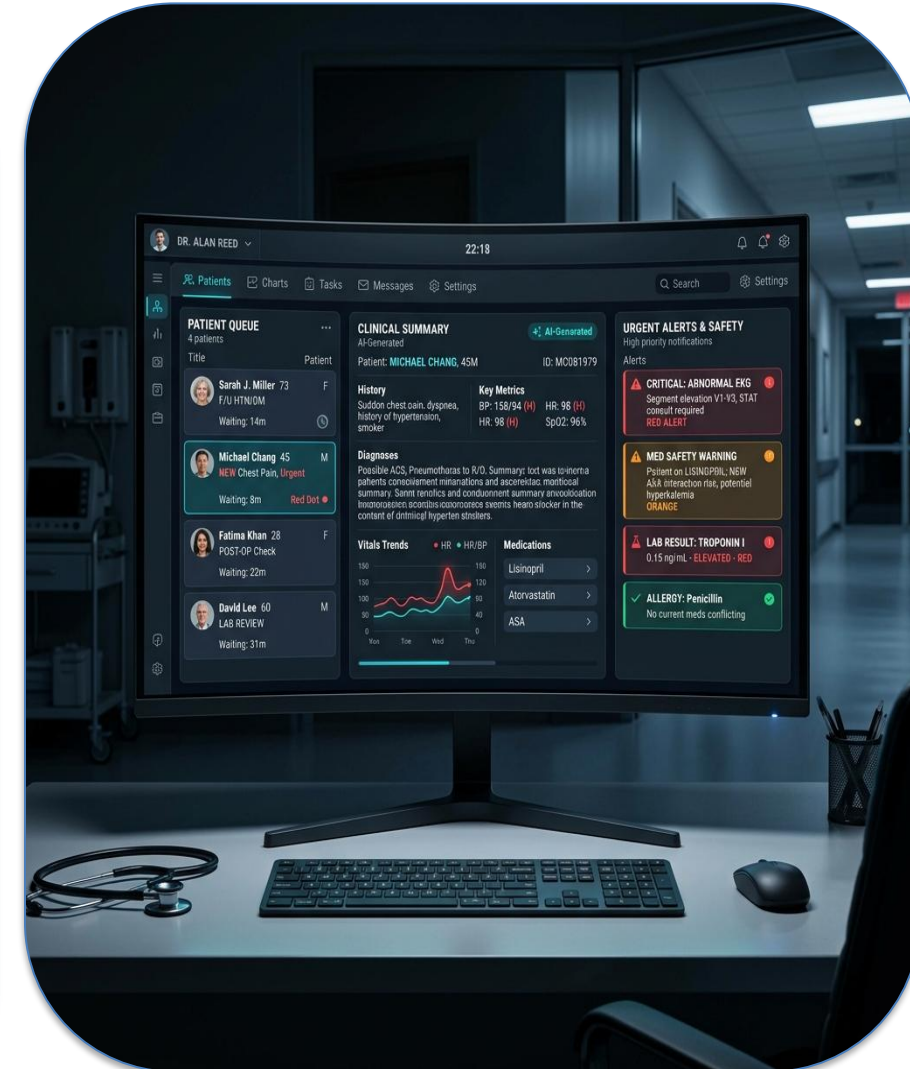
- Native interoperability with national health frameworks for seamless, secure data exchange.

Robust Security Architecture:

- Features strict role-based access, patient-context privacy isolation, and progressive security delays against unauthorized access.

Longitudinal Tracking:

- Versioned clinical analyses and document history ensure that previous AI analyses are preserved and never overwritten.



For Patients:

- In-app notifications, secure access to doctor-approved care info, and explicit sharing controls.

For Nurses:

- Dedicated dashboards with queue management and vital alerts.

For Doctors:

- A real-time clinical copilot that summarizes history and checks for medication conflicts, drastically reducing administrative burden.

For Hospitals:

- Operational oversight and secure backend-to-backend data sharing without direct AI-to-AI communication.

Critical Safety Principle:

- M.I.T.R.A is an assistant, not an autonomous doctor. AI detects concerns, but human professionals review and certify all final decisions.

- Rauf et al. (2017) — *BMJ Open*
- ABDM / NHA — Ayushman Bharat Digital Mission framework, ABHA & FHIR interoperability documentation : <https://abdm.gov.in>
- AI4Bharat / Bhashini — Indian language ASR models : <https://bhashini.gov.in>
- MoHFW — National Digital Health Blueprint, 2019
- Digital Personal Data Protection Act, 2023 : <https://meity.gov.in>